MIL-P-48071A(AR)
15 September 1980
SUPERSEDING
MIL-P-48071(MU)
9 May 1972

MILITARY SPECIFICATION
PROJECTILE (ALTERNATE)
FOR
CARTRIDGE, 40MM, PRACTICE M385
AND
CARTRIDGE, 40MM, HIGH VELOCITY, DUMMY
PARTS FOR

This specification is approved for use by the U.S. Army Armament Research and Development Command and is available for use by all Department and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of projectile designated as Projectile (alternate).

2. APPLICABLE DOCUMENTS

2.1 Issue of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-A-48078	-	Ammunition, Standard, Quality Assurance Provisions General Specification For

FSC: 1310

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1169	-	Packaging, Packing and Marking for Shipment of Inert Ammunition Components

DRAWINGS (See 6.6)

U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

9206916	-	Alternate Projectile For Cartridge, 40MM, Practice, M385

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Parts. The parts shall comply with all requirements specified on Drawing (Dwg) 9206916, all associated drawings and with all requirements specified in applicable specifications.

3.3 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.4 Workmanship. All parts and assemblies shall be fabricated, loaded and assembled in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust, and other foreign matter. The cleaning method used shall not be injurious to any parts, nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous; however, a few light scratches not exposing base material may be permitted. All required marking and stamping shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

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4.2 Classification of inspection. The following types of inspection shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First article inspection. (See 3.3)

4.3.1 Submission. The contractor shall submit a first article as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawings	Quantity
Projectile	9206916	25
Band	9206920	10

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I. FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Projectile and Components				SEE BELOW
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Band (Dwg. 9206920)				
	Examination for defects	25			4.4.2.1
	Projectile (Dwg. 9206916)				
	Examination for defects	25			4.4.2.2
	Push test of rotating band	10			4.5.2

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.1	Band		9206920
			NEXT HIGHER ASSEMBLY 9206916

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR:	None defined				
MINOR					
201.	Number of grooves incorrect		0.65%	3.2	Visual
202.	Evidence of poor workmanship		0.65%	3.4	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48071A(AR)

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
4.4.2.2	Projectile		9206916
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined	see 4.5.1			
MAJOR					
101.	Location of basic diameter on spherical radius		0.40%	3.2	Gage
102.	Concentricity of indicated point on spherical radius with body diameter		0.40%	3.2	Gage
103.	Location of basic diameter on taper		0.40%	3.2	Gage
104.	Concentricity of indicated point on taper with body diameter		0.40%	3.2	Gage
105.	Body diameter		0.40%	3.2	Gage
106.	Total length		0.40%	3.2	Gage
107.	Distance from nose to back of rotating bands		0.40%	3.2	Gage
108.	Location of bourrelet		0.40%	3.2	Gage
109.	Diameter of bourrelet		0.40%	3.2	Gage
110.	Profile of rotating bands		0.40%	3.2	Gage
111.	Concentricity of rotating bands with bourrelet diameter and body diameter		0.40%	3.2	Gage
112.	Diameter between rotating bands		0.40%	3.2	Gage
113.	Diameter of large groove		0.40%	3.2	Gage
114.	Diameter of small groove		0.40%	3.2	Gage
115.	Weight		0.40%	3.2	Balance
116.	Diameter of rotating bands		0.40%	3.2	Gage
117.	Push test of rotating bands	50	0.25%	3.2	4.5.2

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48071A(AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 9206916
4.4.2.2	Projectile				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
MINOR					
201.	Incorrect color		0.65%	3.2	Visual
202.	Marking misleading or unidentifiable		0.65%	3.2	Visual
203.	Flash on rotating bands		0.65%	3.2	Visual
204.	Evidence of poor workmanship		0.65%	3.4	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection Tables.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in Paragraph Reference/Inspection Method column in the tables starting with Paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Test methods and procedures.

4.5.1 Dimensional control of molded or plastic parts. In place of the normal sampling and inspections associated with the Classification of Defects, and after a curing time for the parts has been approved (See Note 1) a sample of three (3) parts (as molded) from each cavity shall be fully inspected dimensionally to qualify a new or reworked cavity for use in production. The molded parts shall carry the individual cavity identification. As a control of each cavity during production, the above quantity of parts from each cavity shall be inspected for at least the defects listed in paragraph 4.4.2.1 after continuous production of each 5,000 parts or at the end of each week, whichever occurs first. Of the three (3) samples, one (1) sample shall be the last part produced. If any defective parts are found during qualification of the cavity, the cavity producing the defective part will not be used in production. If any defective parts are found when inspection is performed for the control of the cavity, the cavity producing the defective part shall be removed from production. Further, that portion of production since the last control check shall be returned to the contractor for inspection for each separate type of defect according to MIL-STD-105, using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. All cavities removed from production because of some fault, may, after reworking, be returned to production providing they pass the qualification test above. The contractor may request a change of inspection frequency providing he presents objective evidence to the Contracting Officer to substantitate the request. Contractor designs of gages and test equipment required to perform the inspections listed herein shall be forwarded to Commander, ARRADCOM, Dover, NJ 07801, ATTN: DRDAR-QAA-I for approval prior to manufacture of equipment. The noted sub-paragraph identify those items and minimum inspections subject to the requirements of this paragraph.

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NOTE 1: In establishing a curing time, dimensionally check ten (10) parts from each cavity at periodic intervals (e.g. every 30, 60, etc mins) until dimensional stability is attained. The curing time will be from the time the part comes out of the mold until dimensional stability is attained. The inspection data used in determining the curing time shall be sent to ARRADCOM, Dover, NJ 07801, ATTN: DRDAR-QAA-S. If there is a change in material, or in the cycle time, or if a cavity is reworked, or a new cavity is used, a new curing time shall be established and approved.

4.5.2 Push test of rotating bands. The projectile shall be placed in an approved fixture and the force specified on the applicable drawing shall be applied. Observation shall be made for the requirement specified on the drawing. This test is a destructive test. Parts so tested shall not be returned to the lot.

NOTE: Movement as specified on Dwg. 9206916 is defined as movement of the back face of the rotating band relative to the body.

5. PACKAGING

5.1 Preservation and packaging.

5.1.1 Level C. Preservation and packaging shall be in accordance with MIL-sTD-1169.

5.2 Packing.

5.2.1 Level C. The projectile, preserved and packaged as specified in 5.1.1, shall be packed in accordance with MIL-STD-1169.

5.3 Marking. Marking shall comply with MIL-STD-1169.

5.4 Shipping. When projectiles from more than one lot are shipped as a carload, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Intended use. The component covered by this specification is intended for use on Cartridge, 40MM, Practice, M385 and Cartridge, 40MM, High Velocity, Dummy.

6.2 Ordering data. See MIL-A-48078.

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6.3 Sumbission of inspection equipment designs for approval. See MIL-A-48078. Submit equipment designs as required to Commander, ATTN: DRDAR-QAA-I, ARRADCOM, Dover, NJ 07801. Request letter of submittal state contractor, contract number, specification number, item nomenclature and classification of defects or test paragraph.

6.4 Submission of results of contractor - conducted examinations and tests. Unless otherwise specified by the Contracting Officer, the contractor shall forward requested records of examinations or tests to Commander, ARRADCOM, ATTN: DRDAR-QAA-S, Dover, NJ 07801.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

6.6 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

Custodian:
Army-AR

Preparing Activity:
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Project Number: 1310-A 247

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